Open-Source Pulse: Theme, Dataset Metrics and Big Data Technique Selection

A follow-up to the lean plan, written after reviewing the Overview page screenshot. Prepared October 22, 2026.

1. Why the current site reads as AI-made

Writers on this topic agree on a few giveaways:

- **Generic copy.** Copy that could describe any product is called the strongest tell, and it has nothing to do with design. A real screenshot of your own interface cannot look generic.
- **Default typeface and palette.** Inter is described as the single strongest tell, and AI tools keep reaching for the same emerald green. Tailwind-default palettes are another giveaway.
- **Everything in rounded cards.** Wrapping every section in a rounded rectangle is a common sign of a vibe-coded site. Layouts built from a grid of equal cells give the same impression.
- **The template layout.** Models default to the statistically common structure: hero, feature cards, then a call to action.
- **Round numbers.** AI tends toward round numbers.
- **Dark mode by inversion.** AI builds a light mode and then inverts it for dark.

The current Overview page hits most of these: a near-black background with a green accent, a hero with four equal cards, a "Best Performance" pill with a glowing card, a generic headline, and round figures (40.0%, 80.0%, 390 Million).

Check the numbers. PR-AUC 0.614 is the same figure that was typed into the earlier README and report, and the generated-at stamp is exactly 23:45:00Z on the last day of the window. If these come from a real run's manifest, that is fine. If they were typed into the page, regenerate them from the pipeline.

2. Theme: "Breakout Forecast desk"

The idea is a weather-forecast bulletin. GH Archive files are the observations, the model issues the forecast, and the forecast is checked afterwards. These terms are real forecasting practice, not decoration:

- The growth-ratio baseline is a **persistence forecast**.
- The base rate is the **climatology**.
- The improvement over the baseline is a **skill score**.
- Calibration is a **reliability diagram**.
- Precision@k with lead time is **forecast verification**.

It is unusual for a project site, and it explains the method.

Visual rules

- **Light first, paper look.** Warm off-white background (#F3EEE3), dark ink text (#1B1F2A), and one alert colour, vermilion (#C8431D), used only for flagged repos. A slate blue (#2F4B6E) serves as the secondary data colour. No gradients, glows or green.
- **Type.** A serif for headlines and body (Newsreader or Source Serif 4), a plain sans for small labels, and a mono face only for numbers in tables. Avoid Inter.
- **Layout.** A single asymmetric column with a narrow margin for side notes, like a newspaper or notebook. Use hairline rules and whitespace instead of cards. Tables are ruled, with right-aligned numbers.
- **Charts.** Thin ink lines, direct labels instead of legends, one annotated peak per chart, and small multiples for growth curves.
- **Copy.** Headlines state a finding written from the real run, for example: "Of N repos created in Q1 2024, M reached 500 stars within four weeks of passing 20." Use exact figures, not rounded ones.
- **Page names.** Overview, Pipeline, Benchmarks, Network, Forecast. This is the same five-page structure as before, with plainer labels.

3. The dataset ledger and findings to compute

These numbers have to come from your own run. `08_export.py` should compute each one in Spark and write it to JSON.

Ledger (keeping track of every file)

- **Per-file record** of all 2,184 hourly files: rows, bytes, parse time and malformed rows per file. Bronze rows must equal the sum.
- **Anomalous hours:** z-score of the hourly event count over a rolling window, to flag outage or spike hours.
- **Event-type mix:** the share of each event type in raw events.
- **Bot share:** bot-filtered events divided by all events.
- **Compression ratio:** raw size divided by Parquet size.

Funnel ratios

- **Retention per stage:** rows kept from raw to bronze, silver, repo-week and candidates.
- **Duplicate-star rate:** duplicates removed in silver divided by star events.
- **Breakout rate:** positive labels divided by candidate repo-weeks (the base rate).

Concentration

- **Skew:** the share of events and stars held by the top 1% of repos, plus a Gini coefficient.
- **Unique actors:** `approx_count_distinct` against an exact count, with the error and the runtime.

Growth findings

- **Time to 20 stars, and from 20 to 500:** median and 90th percentile, breakouts vs non-breakouts.
- **Growth archetypes:** K-means on normalised star curves. Show 4 to 6 curve shapes and the breakout rate of each.

Network

- **Community breakout rate:** the share of each PageRank / label-propagation community that broke out, relative to the base rate.

Forecast verification

- **Lift:** precision@k divided by the base rate.
- **Lead time:** weeks between a repo's first flag and its crossing 500 stars.
- **Reliability:** predicted probability bins against the observed breakout rate.

- **Skill score:** improvement over the persistence baseline.

4. Big data ideas considered and selected

Keep (these make up the project)

1. A Parquet lake with an explicit schema, zstd compression and date partitions.
2. Partition pruning and predicate pushdown (B1, B2).
3. Early aggregation and window features at repo-week grain, with the approximate vs exact distinct-count comparison (B3).
4. Adaptive query execution and skew handling (B4).
5. A scaling study across data size and cores (B5).
6. The file ledger with anomaly flags.
7. GraphFrames PageRank and label propagation, built point-in-time, with the degree-cap guard (B6).
8. A MLlib GBT with a time split, compared against the persistence baseline.
9. K-means growth archetypes.
10. Forecast verification (lift, lead time, reliability).

Optional, only if time remains

- Structured Streaming replay of a few hourly files.
- A multi-node Spark cluster.

Drop

Bucketing, Delta or Iceberg, a DuckDB comparison, small-file compaction, extra graph algorithms, and rolling cross-validation. Each costs time and adds little that the selected items do not already show.

Only two things were added to the earlier lean plan: the file ledger and the growth archetypes. Both produce visible findings.

5. Page changes

- **Overview:** one finding headline, then the ledger as a ruled table, with no cards.

- **Pipeline:** the funnel with retention ratios and the per-file ledger, including the anomaly hours.
- **Benchmarks:** B1 to B6 as annotated charts.
- **Network:** communities with their breakout rates.
- **Forecast:** the weekly watchlist with hit and miss outcomes, the reliability chart, the lift, and the archetype small multiples.

Sources on AI-generated design tells

- 21st.dev: "How to Make a Site Not Look AI Generated" (https://21st.dev/blog/website-not-look-ai-generated)
- "How to detect AI slop in your design" (https://github.com/Laith0003/ux-skill/wiki/How-to-detect-AI-slop-in-your-design)
- Shuffle: "Why Do Most AI-Generated Websites Look the Same?" (https://shuffle.dev/blog/2026/01/why-do-most-ai-generated-websites-look-the-same/)
- "How to Stop Your Vibe Coded Website From Looking AI Generated" (https://app.notion.com/p/How-to-Stop-Your-Vibe-Coded-Website-From-Looking-AI-Generated-3a264170dad0804eb2f2ef31591e60ab)
- So Websites: "Publish an AI generated website" (https://www.sowebsites.co.za/blog/publish-ai-generated-website)